

**YEAR 9 SCIENCE
PHYSICAL SCIENCES
LIGHT TEST (MODIFIED)**

NAME: SOLUTIONS

CLASS: _____

Mark: $\frac{\quad}{27}$ **Achievement standards being tested**

Energy transfer through different mediums can be explained using wave and particle models.

Mark	ND	NW	C	HC	O
Mark Range					

Multiple Choice Write the answer to each question in the appropriate box at right.

1. The light energy we feel as heat is:

- (a) infrared.
- (b) ultraviolet.
- (c) microwave.
- (d) x-ray.

2. The part of the eye that an image forms on is the:

- (a) iris.
- (b) lens.
- (c) retina.
- (d) optic nerve.

3. Refraction of light is:

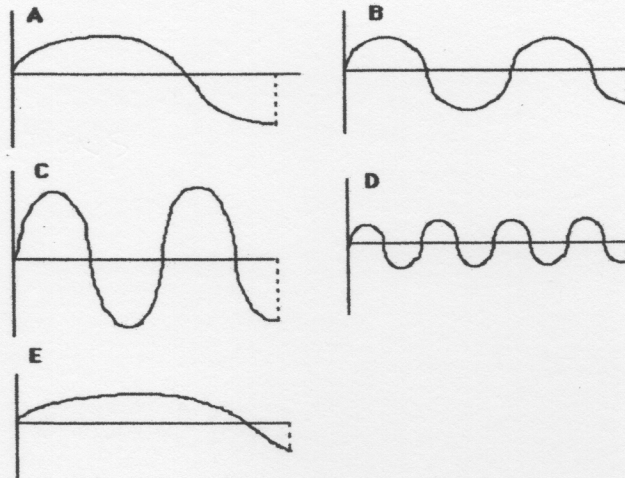
- (a) the wave motion of light as it passes through a substance.
- (b) the bending of light as it passes from one substance into another.
- (c) a reflection of light rays at the surface of a medium.
- (d) the failure of light to pass through a very dense substance.

4. Light travels in _____ lines.

- (a) diagonal
- (b) curved
- (c) horizontal
- (d) straight

QUESTION	ANSWER
1	A
2	C
3	B
4	D
5	C
6	D
7	C
8	A

Questions 5 refers to the following diagrams.



5. The wave with the greatest **amplitude** is:

- (a) A
- (b) B
- (c) C
- (d) D
- (e) E

6. Mirrors will _____ light rays.

- (a) scatter
- (b) refract
- (c) transmit
- (d) reflect

7. Which material can be classified as **translucent**, allowing some light through?

- (a) A piece of cardboard.
- (b) Window glass.
- (c) Frosted glass.
- (d) Carpet.

8. The speed of light in air is:

- (a) 300 000 km/sec.
- (b) 30 km/sec.
- (c) 340 km/sec.
- (d) 3000 km/sec.

Short Answer

Write the answer to the questions in the spaces provided.

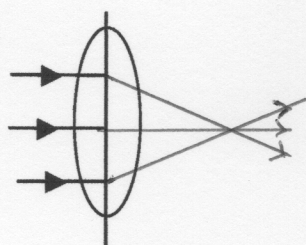
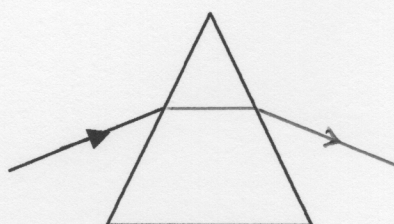
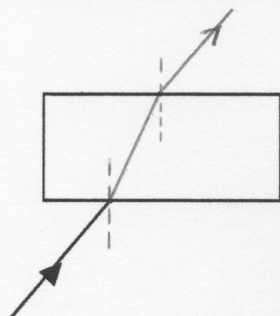
- Light is a **transverse wave**. In which direction do the particles move?
(Circle the correct answer.)

Up and down.

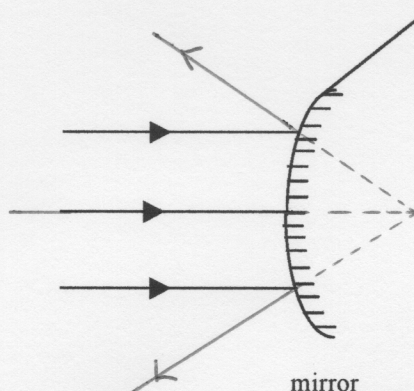
Backwards and forwards (left and right).

(1)

- Draw the path of the rays in the following diagrams.

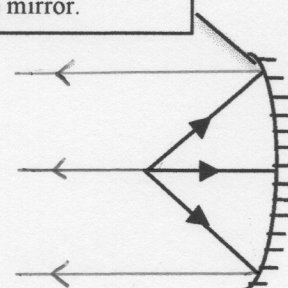


lens



mirror

The shiny surface is at the front of the mirror.



mirror

[1 mark each.]

(5)

- (a) Label the following diagram of a wave.

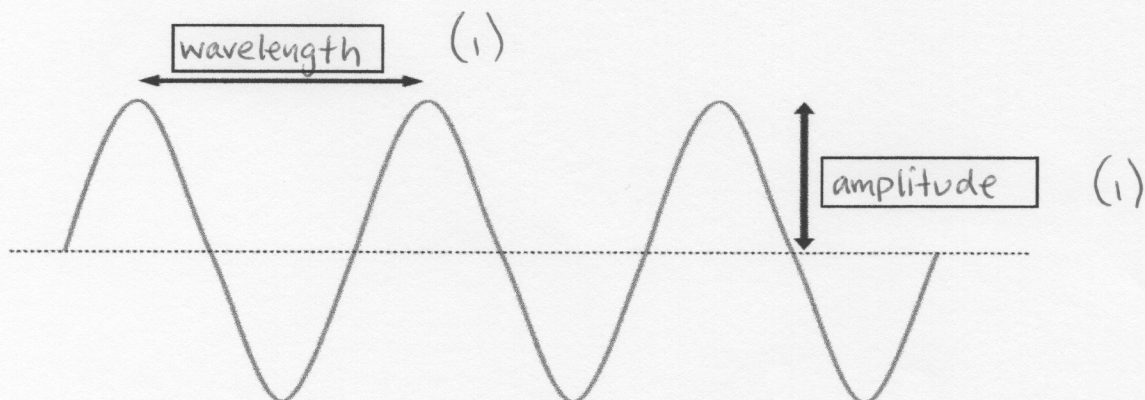
Choose from the following words.

frequency

speed

amplitude

wavelength



(2)

- (b) Complete the diagram of the electromagnetic spectrum by naming the remaining bands.

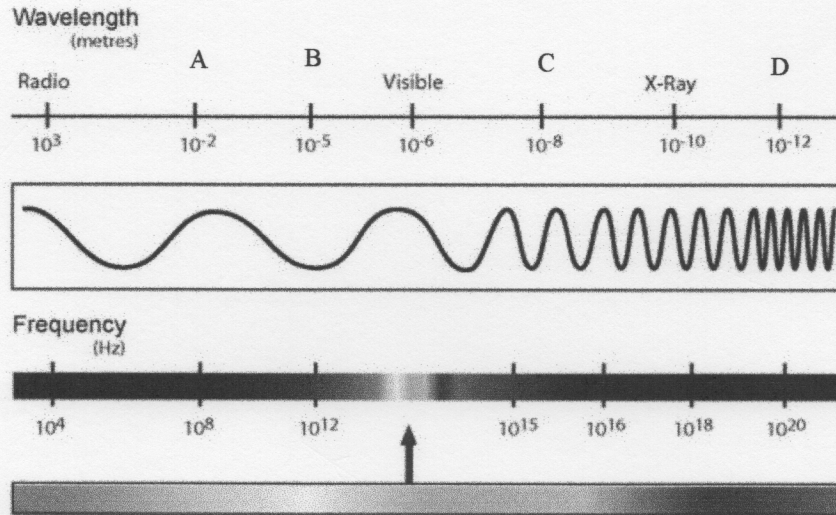
Choose from the following words.

gamma ray

microwave

ultraviolet

infrared



A: microwave

B: infrared

C: ultraviolet

D: gamma ray

(4)

4. (a) Label the following illustration of the human eye. Choose from the following words.

iris

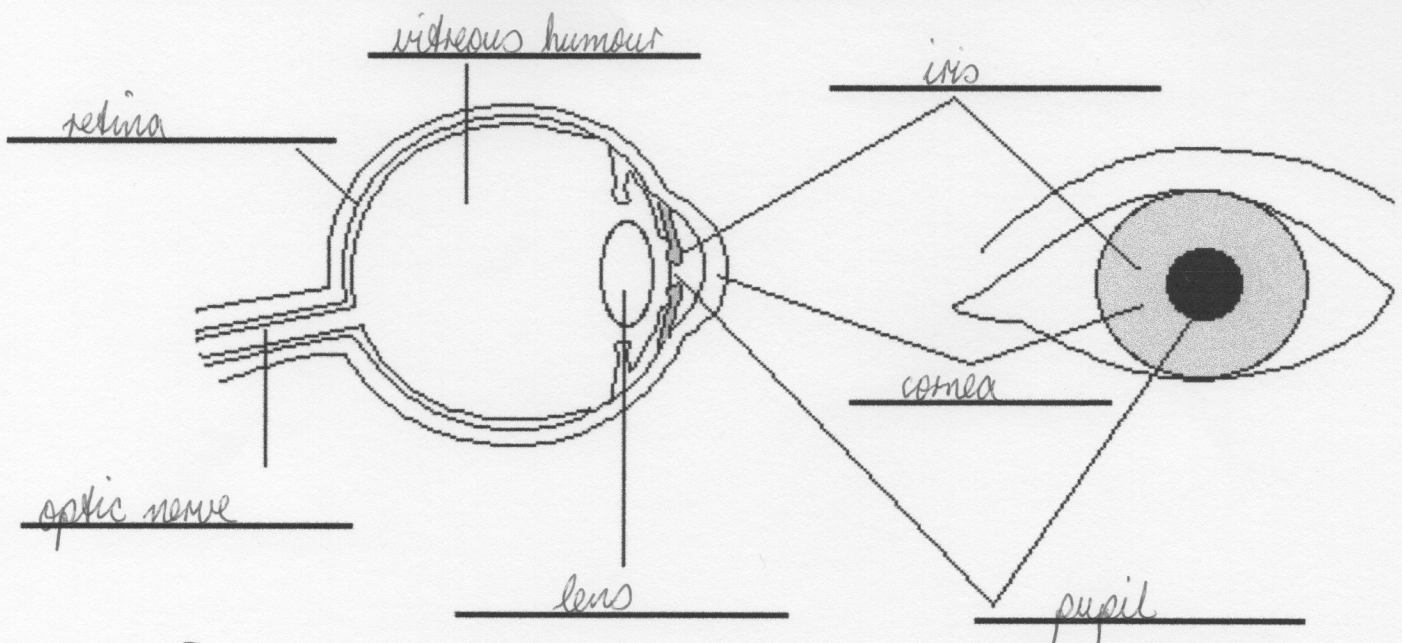
optic nerve

retina

pupil

lens cornea

vitreous humour



[1 mark each.]

(7)